

```
# =====
```

```
# Assignment 1
```

```
# Name: Abdulkadir Musa
```

```
# =====
```

```
# -----
```

```
# Part A: Variables & Data Types
```

```
# -----
```

```
# Creating variables
```

```
name = "Abdulkadir Musa"    # String
```

```
age = 22                    # Integer
```

```
cgpa = 4.50                 # Float
```

```
is_python_student = True    # Boolean
```

```
# Printing data types
```

```
print("Part A")
```

```
print(type(name))
```

```
print(type(age))
```

```
print(type(cgpa))
```

```
print(type(is_python_student))
```

```
# -----
```

```
# Part B: Lists
```

```
# -----
```

```
# Creating a list of gadgets
```

```
gadgets = ["Laptop", "Smartphone", "Tablet", "Headphones", "Smartwatch"]
```

```
print("\nPart B")
```

```
# Print first and last gadget
```

```
print("First gadget:", gadgets[0])
```

```
print("Last gadget:", gadgets[-1])
```

```
# Add two more gadgets
```

```
gadgets.append("Camera")
```

```
gadgets.append("Power Bank")
```

```
# Remove one gadget from the middle
```

```
gadgets.pop(3)
```

```
# Print final list length
```

```
print("Final list:", gadgets)
```

```
print("Length of list:", len(gadgets))
```

```
# -----
```

```
# Part C: Tuples
```

```
# -----
```

```
# Creating a tuple
```

```
coordinates = (10, 20, 30)
```

```
print("\nPart C")
```

```
# Tuples are immutable.
```

```
# Uncommenting the next line will raise an error:
```

```
# coordinates[1] = 99
```

```
# Error:
```

```
# TypeError: 'tuple' object does not support item assignment
```

```
# This happens because tuples cannot be changed after creation.
```

```
# Print last item
```

```
print("Last coordinate:", coordinates[-1])
```

```
# -----
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```
# Part D: Dictionaries
```

```
# -----
```

```
student = {  
    "name": "Abdulkadir Musa",  
    "matric_no": "SLUK/22/IT/001",  
    "department": "Information Technology",  
    "age": 22  
}
```

```
print("\nPart D")
```

```
# Print name and department
```

```
print("Name:", student["name"])
```

```
print("Department:", student["department"])
```

```
# Add CGPA
```

```
student["cgpa"] = 4.50
```

```
# Update age
```

```
student["age"] = 23
```

```
# Print keys and values
```

```
print("Keys:", student.keys())
```

```
print("Values:", student.values())
```

```
# -----
```

```
# Part E: Mixed Practice
```

```
# -----
```

```
class_list = [
```

```
    {"name": "Abdulkadir", "score": 85},
```

```
    {"name": "Aisha", "score": 90},
```

```
    {"name": "Musa", "score": 88}
```

```
]
```

```
print("\nPart E")
```

```
# Print second student's name
```

```
print("Second student:", class_list[1]["name"])
```

```
# Print last student's score
```

```
print("Last student's score:", class_list[-1]["score"])
```